

Project Cheap Adapter: one world, different coordinates

A bounded test of the Platonic Representation Hypothesis, converted into deployable linear adapters and a shared multi-encoder hub.

Project Documents

34 verdicts | 21 confirmed | 7 falsified | 4 inconclusive | 2 not meaningful

EXECUTIVE FINDING

The project confirms PRH in a narrow form: trained independent models share mainly local structure that linear maps make useful. It also bounds the claim: convergence is not global or universal, and visible geometry need not transfer.

RESEARCH PROGRAM

8 encoders | 2 shipped systems

1

Experiment A

GPT-2 vs Pythia-160M

CKA + layer-wise ridge on 10,000 passages tests whether independent language models share linearly accessible structure.

R² mean 0.737; peak 0.797

2

Experiment B

MobileCLIP to SigLIP 2

One 512x768 ridge matrix aligns phone and server image encoders and is exported as a 0.79 MB fp16 Core ML artifact.

R@5 = 95.3% of native ceiling

3

Cheap Adapter hub

Seven experts, one space

One entry map per encoder reaches a shared 512-d whitened hub, replacing quadratic pairwise translation with linear growth.

N maps, not N(N-1)

EVIDENCE LADDER - STRONGER CLAIMS REQUIRE STRONGER TESTS

1	Shared neighbours	mutual k-NN	strong locally
2	Shared order	distance-rank rho	0.29 cross-modal
3	Shared structure	CKA / RSA	partial
4	Shared subspace	ridge + retrieval	88-97% of ceiling
5	Same geometry	rotation only	rejected: 0.038

The pattern is the result: models agree most on local neighbourhoods, less on global shape, yet become highly interoperable after a fitted linear stretch.

HEADLINE EVIDENCE

0.9183

MLP matches ridge in tested setup

0.592

vision adapter held-out R²

0.768

bge-SBERT raw shape rho

96.7%

ConvNeXt hub transfer / native

RELATION TO HUH ET AL.

Huh et al. linked alignment to scale and capability. Cheap Adapter tests the narrower mechanism: learned local structure is linearly usable across modalities and lineages, with held-out ConvNeXt at 96.7% of native. Limits: no universal or global identity; scaling is not monotonic beyond the tested range; the source-rank cliff is numerical.

From one adapter to the Cheap Adapter hub

The same scientific idea becomes an architecture: freeze every encoder, fit one linear entry map per space, and reuse downstream components across models.

SYSTEM DESIGN

PAIRWISE

7 encoders -> 42 ordered maps

Adding one model requires maps to and from every existing model.

CHEAP ADAPTER HUB

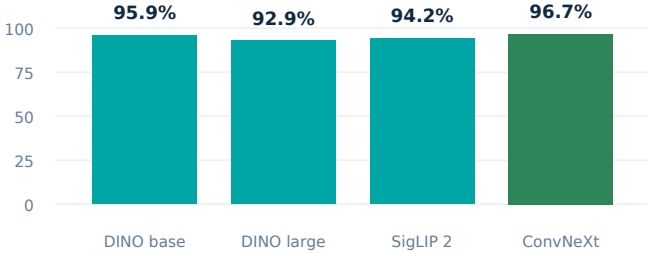
7 encoders -> 7 entry maps

DINO-small trains head once; ConvNeXt fits map only



MEASURED TRANSFER - HELD-OUT 1,000-ITEM BGE-M3 RETRIEVAL

% of a native bge-m3 retrieval head



WHAT TRANSFERRED

- The caption head is trained once from DINOv2-small hub vectors to bge-m3 text targets.
- ConvNeXt is external: only its entry map is fitted; the hub and caption head stay frozen.
- ConvNeXt reaches 96.7% of native bge-m3 retrieval; adaptation, not zero-shot.

THREE COMPLEMENTARY RESULTS

MAP-FREE

rho = 0.290

Mean cross-modal distance-rank agreement before any map; bge-SBERT reaches 0.768.

COMPOSED

103.6%

Mean hub performance versus bespoke pairwise maps across all 42 ordered pairs.

DEPLOYED

0.79 MB

The fp16 MobileCLIP-to-SigLIP matrix reaches 95.3% of native R@5.

WHY THE INFERENCE IS CREDIBLE

Frozen models only closed-form maps are fitted	Held-out scoring 8,533 train / 1,000 test in the hub	Negative controls shuffle and random maps sit at chance	Nonlinear control MLP ties linear at 0.9183
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PROJECT IN ONE PARAGRAPH

Two models do not share coordinates: direct cross-space retrieval is at chance without an adapter. Yet they encode related structure, and one phone-sized matrix makes it usable. What they share most strongly is local - nearby items agree more than the global shape. Coordinate frames are cheap to fix; semantic arrangement is what training buys. This does not imply one identical latent space.

Fractal probe: the role of training data

FractalDB-only encoders provide a sharp control: structured training with zero natural photographs, bracketed by matched random-init and ImageNet twins.

THREE TESTS, ONE CONSISTENT READING

G8

Visible geometry

Four map-free metrics reproduce nonzero cross-modal structure. Raw shape rho is 0.288 (0.329 without collapsed GPT-2); whitening lowers it to 0.062.

G9

Fractal vs controls

P1/P2 confirm real geometry above matched random controls; P5w confirms after whitening. The CNN signal survives PC1 removal; P3 asymmetry remains inconclusive.

G10

Functional recovery

H1 is falsified: fractal transfer stays at the floor and matches random twins. H3 CNN/ViT ratios are not meaningful under the floor guard.

G9 - VISIBLE STRUCTURE RELATIVE TO MATCHED CONTROLS

Reality fraction R

0 = random architecture; 1 = ImageNet twin

Fractal CNN

0.09

Fractal ViT

0.21

Matched-norm CNN control: 0.067 fractal vs 0.036 random. The training effect survives preprocessing correction.

Robustness and uncertainty

- CNN image-vs-text R gap: 95% CI [-0.106, +0.100].
- ViT image-vs-text R gap: 95% CI [-0.053, +0.135].
- Corrected drop-1: CNN ordering survives; ViT falls below random.
- CNN-vs-ViT contrast is not interpreted because capacity differs.

G10 - FROZEN HUB/HEAD RETRIEVAL (R@1; OPERATIONAL FLOOR = 0.050)

ENCODER	TRANSFER	NATIVE	RANDOM TWIN	INTERPRETATION
Fractal CNN	0.014	0.014	0.012	at floor; no training-specific gain
Fractal ViT	0.020	0.022	0.017	at floor; no training-specific gain
ImageNet CNN	0.258	0.274	-	clear functional transfer
ImageNet ViT	0.204	0.208	-	clear functional transfer

PROJECT VERDICT LEDGER - 34 PRE-REGISTERED PREDICTION UNITS

21

CONFIRMED

Includes learned geometry, cross-modal and cross-lineage transfer, and G9 P1/P2/P5w.

7

FALSIFIED

Includes H1 semantic recovery through the hub; it does not mean that all fractal effects failed.

4

INCONCLUSIVE

P3 image-vs-text reality asymmetry: two architectures across raw and whitened views.

2

NOT MEANINGFUL

H3 CNN and ViT percentage-of-native ratios are blocked by the 0.050 floor guard.

HONEST HEADLINE AND SCOPE

Confirmed narrowly: trained independent models share mainly local structure that linear maps make useful across modalities and lineages. Bounded: not global or universal; scale is not monotonic, natural-world data matters, and visible geometry need not transfer.